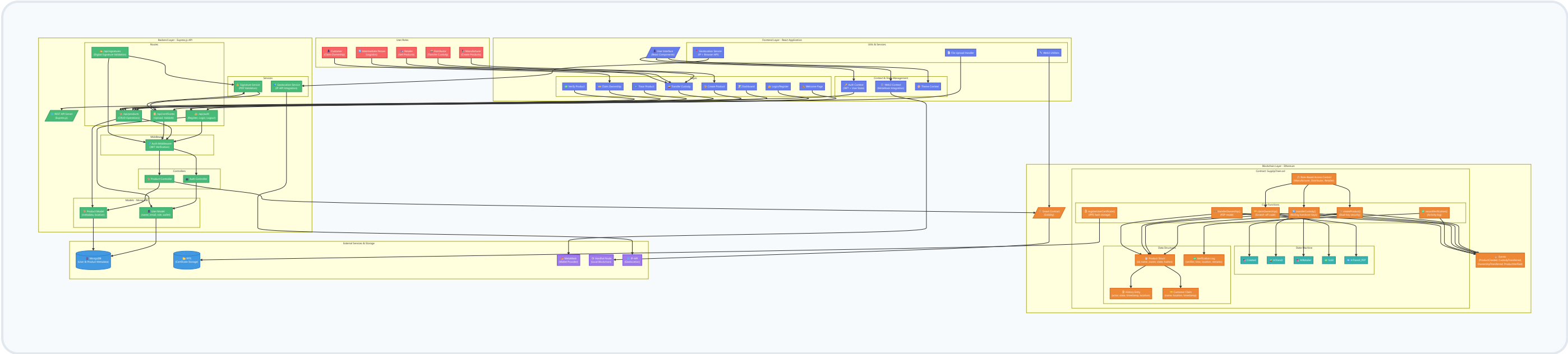


Blockchain Supply Chain System

Comprehensive System Architecture Diagram



Frontend Layer

React components, pages, context, and utilities

Backend Layer

Express.js API, routes, controllers, and services

Blockchain Layer

Solidity smart contracts and blockchain logic

Data Storage

MongoDB database and IPFS distributed storage

External Services

MetaMask, Hardhat node, and third-party APIs

User Roles

Different actors in the supply chain system

Technology Stack

Frontend

Backend

Database

Blockchain

Storage

IPFS (Certificate Storage)

Security

React.js, Web3.js,
MetaMask Integration

Node.js, Express.js, JWT
Authentication

MongoDB (User & Product
Metadata)

Ethereum, Solidity ^0.8.20,
Hardhat

RBAC, JWT, Digital
Signatures, Dual-Key
System

External APIs

IP-API (Geolocation),
Browser Geolocation API

Development

Hardhat, OpenZeppelin,
Bun/NPM